OLG with Production

Diamond Framework

Contents

From Exchange to Production

What Changes When We Add Production?

Lecture 1 recap

Each gen t solves max $U(c_t^1)+\beta U(c_{t+1}^2)$ s.t. $c_t^1+s_t=w_1$, $c_{t+1}^2=(1+r_{t+1})s_t$, with **fixed** w_1, w_2 and r_{t+1} from $s_t = m/p_t$.

Three things change with production:

1. **Endowments endogenous:** $w_t = \omega(k_t) \equiv f(k_t) - k_t f'(k_t)$ (competitive wage).
2. **Interest rate from capital:** $r_{t+1} = \rho(k_{t+1}) \equiv f'(k_{t+1}) - \delta_K$, not from money. Savings $\to$ physical capital k_{t+1}.
3. **Capital market clears:** $(1+n)k_{t+1} = s_t$. Transition map $k_{t+1} = g(k_t)$ replaces the offer-curve cobweb algorithm.

What stays the same

The Euler equation $U'(c_t^1) = \beta(1+r_{t+1})U'(c_{t+1}^2)$ is identical. The savings function $s(w,r)$ is Lecture 1's $f(r)$ with endogenous (w,r). All FWT pathologies of Lecture 1 persist — in a new form.

Lecture 1 – 2: Connections

Concept	L1 (exchange)	L2 (production)
Young's endowment	w_1 (exogenous)	$w_t = \omega(k_t)$ (endogenous)
Return on savings	r (money market)	$r_{t+1} = f'(k_{t+1}) - \delta_K$
Savings vehicle	Fiat money / bonds	Physical capital k_{t+1}
Market clearing	$s_t = m/p_t$	$(1+n)k_{t+1} = s_t$
Equilibrium dynamics	Cobweb algorithm	Transition map $g(k_t)$
Samuelson condition	$\bar{r} < 0$	$r^* < n$ (dyn. inefficiency)
Social transfer	Outside money	Capital / PAYG
FWT failure	$\sum_t p_t \bar{e}_t = +\infty$	$k^* > k^{\mathrm{GR}}$
Pareto improvement	PAYG / public debt /money	PAYG / public debt /money
Golden Rule	$r^* = 0$: budget$\equiv$feasibility	$r^* = n$: budget slope$= -(1+n)$

Every row is a strict generalisation. Setting $f(k) = w_1 + (1+r)k$ (linear) reduces the production model to L1 exactly.

Households

Household Problem (Diamond 1965)

Preferences: $U_t = u(c_t^1) + \beta\, u(c_{t+1}^2)$, $\beta > 0$, $u' > 0$, $u'' < 0$, Inada.

Budget constraints: $c_t^1 + s_t = w_t$ (young), $\quad c_{t+1}^2 = (1 + r_{t+1})\, s_t$ (old).

Euler equation (FOC):

$$\boxed{u'(c_t^1) = \beta\,(1 + r_{t+1})\, u'(c_{t+1}^2)} \quad \Rightarrow \quad s_t = s(w_t,\, r_{t+1}).$$

Existence and Uniqueness of Optimal Savings

Proposition

For any $w_t > 0$, $r_{t+1} > -1$: $\exists!\ s_t \in (0, w_t)$ satisfying the Euler equation. Write $s_t = s(w_t, r_{t+1})$.

Proof.

Define $\Phi(s) \equiv u'(w_t - s) - \beta(1 + r_{t+1})u'((1 + r_{t+1})s)$.

Existence. $\Phi(0) = u'(w_t) - \beta(1 + r_{t+1}) \cdot u'(0^+) = -\infty < 0$ (Inada: $u'(0) = +\infty$). $\Phi(w_t) = u'(0^+) - \beta(1 + r_{t+1})u'((1 + r_{t+1})w_t) = +\infty > 0$. IVT: $\exists\, s^* \in (0, w_t)$ with $\Phi(s^*) = 0$.

Uniqueness. $\Phi'(s) = -u''(w_t - s) - \beta(1 + r_{t+1})^2 u''((1 + r_{t+1})s) > 0$ since $u'' < 0$. Φ strictly increasing $\Rightarrow$ at most one zero. $\square$

The Savings Function: Comparative Statics

From implicit differentiation of $\Phi(s) = 0$:

Wage effect: $\dfrac{\partial s}{\partial w} = \dfrac{-u''(c^1)}{-u''(c^1) - \beta(1+r)^2 u''(c^2)} \in (0,1)$. Savings always increase in w.

Interest-rate effect: $\sigma(c) \equiv -u''(c)c/u'(c) > 0$ (inverse IES ζ).

Proposition

$$\frac{\partial s}{\partial r} = \frac{\beta u'(c^2)\left[1 - \sigma(c^2)\right]}{-u''(c^1) - \beta(1+r)^2 u''(c^2)}.$$

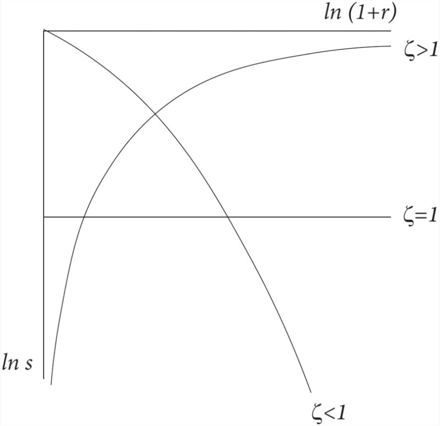
Denominator > 0, so $\partial s/\partial r \gtrless 0 \iff \sigma(c^2) \lessgtr 1 \iff \text{IES} \gtrless 1$.

Connection to Lecture 1: backward-bending offer curve

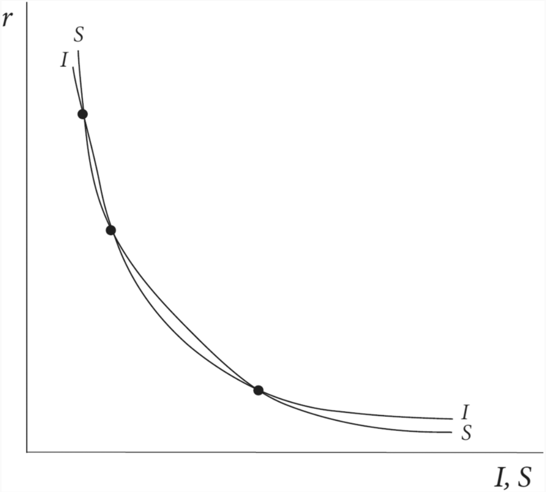
In L1 the offer curve bends backward when $\partial s/\partial r < 0$, i.e.$\zeta < 1$ ($\sigma > 1$) — generating endogenous cycles. Here *exactly the same condition* drives **poverty traps** and non-monotone transition maps in L2.

Log Utility: Closed Form

With $u(c) = \ln c$: $\sigma = \zeta \equiv 1$, $\partial s/\partial r = 0$. From $1/(w-s) = \beta(1+r)/[(1+r)s]$: $\boxed{s_t = \frac{\beta}{1+\beta} w_t}$ — fixed fraction of wage.



Multiplicity in I-S Framework



Production Sector and Factor Markets

Production Technology

CRS production: $Y_t = F(K_t, N_t)$, per worker:

$$y_t = f(k_t), \quad f' > 0,\ f'' < 0,\ f'(0) = +\infty,\ f'(\infty) = 0.$$

Competitive factor prices:

$$w_t = \omega(k_t) \equiv f(k_t) - k_t f'(k_t), \qquad r_t = \rho(k_t) \equiv f'(k_t) - \delta_K.$$

Monotonicity

$\omega'(k) = -kf''(k) > 0$: wages $\uparrow$ in k. $\rho'(k) = f''(k) < 0$: returns $\downarrow$ in k. In L1 the endowment w_1 was *fixed* and r was a parameter. Here both are *moving* functions of k_t.

Capital Market Clearing and the Transition Map

Labour: N_t young workers supply inelastically.
Capital: $K_{t+1} = N_t s_t$.
Dividing by $N_{t+1} = (1+n)N_t$:

$$\boxed{(1+n)\, k_{t+1} = s\big(\omega(k_t),\ \rho(k_{t+1})\big).}$$

Fundamental difference equation of Diamond (1965).

Equilibrium Dynamics and Stability

The Transition Map $k_{t+1} = g(k_t)$

By the IFT, $(1+n)k_{t+1} = s(\omega(k_t), \rho(k_{t+1}))$ implicitly defines g wherever $(1+n) - s_r\rho'(k_{t+1}) \neq 0$. Differentiating:

$$g'(k) = \frac{s_w\, \omega'(k)}{(1+n) - s_r\, \rho'(g(k))}.$$

Local stability

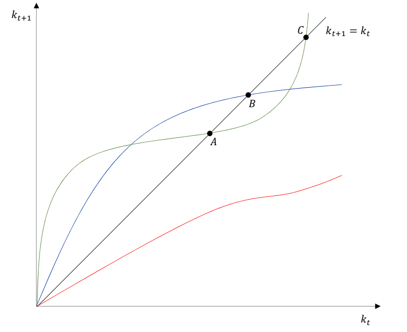
k^* locally stable $\Leftrightarrow |g'(k^*)| < 1$. More likely when: (i) IES≥ 1 ($s_r \geq 0$); (ii) $|f''|$ small; (iii) n large. IES< 1 ($s_r < 0$) can destabilise — same condition as backward-bending offer curve (Lecture 1) that generates cycles.

Steady States: Existence, Uniqueness, Multiplicity

Case 1 — Unique stable k^*
g crosses 45° line once from above. Guaranteed with log utility + CD ($g' = \alpha < 1$).

Case 2 — Multiple steady states
g non-monotone when $s_r < 0$ (IES< 1): two stable $k_L < k_H$, one unstable k_u. Poverty trap: start below $k_u \Rightarrow k_L$. *Production-economy counterpart of L1's multiple equilibria.*

Case 3 — No interior SS
$g(k) < k$ or $g(k) > k$ everywhere.



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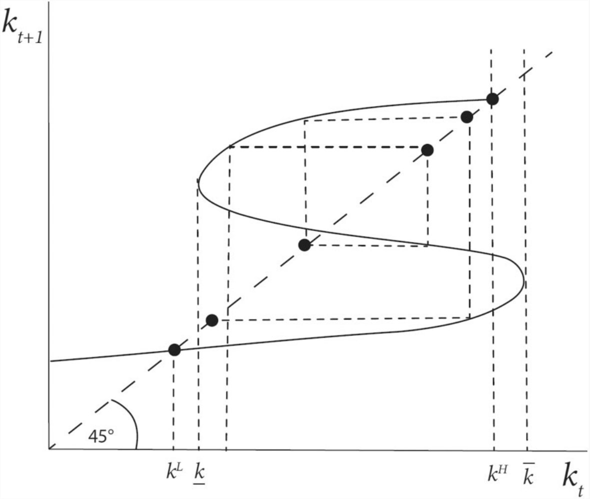


Cobb-Douglas + Log: Closed-Form Solution

Parametric Example: Cobb-Douglas + Log Utility

$f(k) = k^\alpha$, $\omega(k) = (1-\alpha)k^\alpha$,
$\rho(k) = \alpha k^{\alpha-1} - \delta_K$.
Log: $s_t = \frac{\beta}{1+\beta}\omega(k_t)$. Clearing:
$(1+n)k_{t+1} = \frac{\beta}{1+\beta}(1-\alpha)k_t^\alpha$.

Transition map

$$k_{t+1} = \underbrace{\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}}_{A} k_t^\alpha \equiv g(k_t)$$

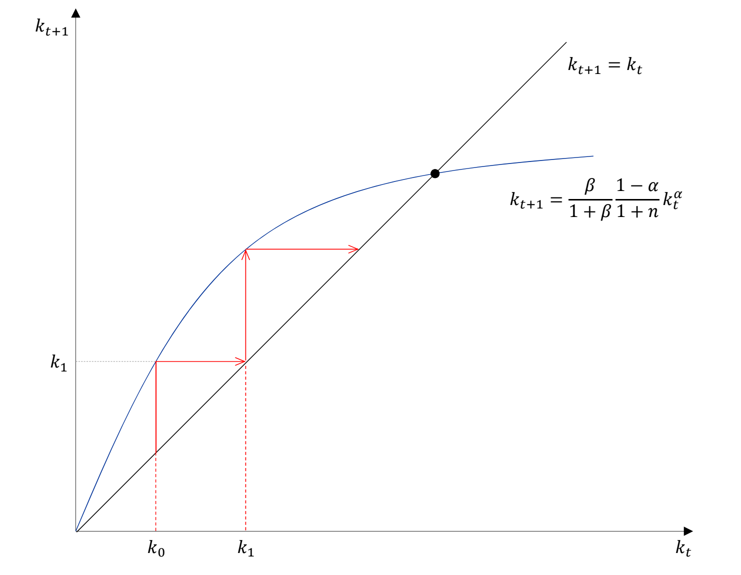
$k^* = A^{1/(1-\alpha)}$. $g'(k^*) = \alpha < 1$. ✓

Calibration ($\delta_K = 0$, period $= 30$ yr):
$\alpha = 0.36$, $\beta \approx 0.40$, $n = 0.10$.

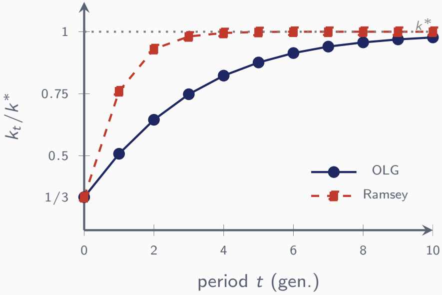
	Formula	Value
A	$\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}$	0.166
k^*	$A^{1/(1-\alpha)}$	0.061
r^*	$\alpha(k^*)^{\alpha-1}$	2.166
n	population growth	0.100

$r^* \approx 2.17 \gg n = 0.10$ (generational)
$\Rightarrow$ $r_{\text{ann}} \approx 3.9\%$, $n_{\text{ann}} \approx 0.32\%$: **dynamically efficient**. To get $r^* < n$ need very high β (patient) or very low α.

Parametric Example: Cobb-Douglas + Log Utility



Transition Dynamics: OLG vs. Ramsey



Both paths start at $k_0 = k^*/3$ and both reach k^*. The difference is *not* the speed of k_t but the **jump in c_t at** $t = 0$: Ramsey yes, OLG no.

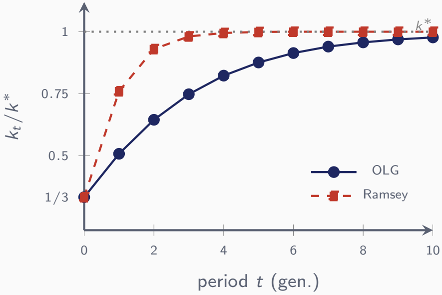
Ramsey — 2D system in (k_t, c_t):

$$\begin{cases} (1+n)k_{t+1} = f(k_t) + (1-\delta)k_t - c_t \\ u'(c_t) = \beta\left[1 + f'(k_{t+1}) - \delta\right]u'(c_{t+1}) \end{cases}$$

$+$ TVC: $\lim_t \beta^t u'(c_t)k_{t+1} = 0$.

Linearised system has eigenvalues $\mu_1 \in (0,1)$, $\mu_2 > 1$. TVC kills $\mu_2 \Rightarrow c_0$ **jumps** onto saddle path. Persistence of k: μ_1.

Transition Dynamics: OLG vs. Ramsey



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Ramsey — 2D system in (k_t, c_t):

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OLG — 1D system in k_t alone:

$$(1+n)k_{t+1} = s(\omega(k_t)) = \tfrac{\beta}{1+\beta}(1-\alpha)k_t^{\alpha} \equiv g(k_t)$$

No TVC needed. Linearise:

$$k_{t+1} - k^* \approx g'(k^*)(k_t - k^*), \quad g'(k^*) = \alpha.$$

Convergence rate = capital share.

The Command Optimum

Setup: Social Welfare and Resources

Social welfare function — planner weights cohorts by $(1+R)^{-t}$:

$$\mathcal{U} = \beta\, u(c_0^2) + \sum_{t=0}^{T-1} \frac{1}{(1+R)^t} \left[u(c_t^1) + \beta\, u(c_{t+1}^2) \right].$$

R: intergenerational discount rate. $R = 0$ = *equal weights on all cohorts*.

Aggregate resource constraint:

$$K_t + F(K_t, N_t) = K_{t+1} + N_t c_t^1 + N_{t-1} c_t^2.$$

Dividing by N_t and using $N_{t-1} = N_t/(1+n)$:

$$\underbrace{k_t + f(k_t)}_{\text{resources per young}} = \underbrace{(1+n)k_{t+1}}_{\text{cap. for next cohort}} + c_t^1 + \tfrac{1}{1+n} c_t^2.$$

Lagrangian and First-Order Conditions

Form the Lagrangian:

$$\mathcal{L} = \beta u(c_0^2) + \sum_{t=0}^{T-1} \frac{1}{(1+R)^t}\Big\{u(c_t^1) + \beta u(c_{t+1}^2) + \lambda_t\big[k_t + f(k_t) - (1+n)k_{t+1} - c_t^1 - \tfrac{c_t^2}{1+n}\big]\Big\}.$$

Maximising w.r.t. c_t^1, c_{t+1}^2, k_{t+1}:

$$\lambda_t = u'(c_t^1), \tag{1}$$

$$\beta u'(c_{t+1}^2) = \frac{1}{1+R}\frac{\lambda_{t+1}}{1+n}, \tag{2}$$

$$\lambda_t = \frac{1}{1+R}\frac{\lambda_{t+1}}{1+n}\big[1 + f'(k_{t+1})\big]. \tag{3}$$

(1): marginal utility of young = shadow value of resources.
(2): marginal utility of old, discounted by β, equals the shadow price *next period* adjusted by planner discount and cohort size.
(3): Euler for capital — saving one unit today costs λ_t and returns $[1 + f'(k_{t+1})]$ units of resources next period.

Planner's Euler Equation and Steady State

Combining (1)–(3) gives the **planner's Euler**:

$$u'(c_t^1) = \beta\big[1 + f'(k_{t+1})\big]\,u'(c_{t+1}^2).$$

Identical in form to the household Euler — only the *price* of intertemporal trade differs (here $1 + f'(k_{t+1})$ is the social rate of return, not a market r).

Steady state ($c_t^1 = c^1$, $c_t^2 = c^2$, $k_t = k$):

$$\beta u'(c^2) = \frac{1}{1+R}\,\frac{u'(c^1)}{1+n}, \quad 1 + f'(k) = (1+R)(1+n)$$

Expanding $(1+R)(1+n) = 1 + R + n + Rn$ and neglecting Rn:

$$f'(k) \;\approx\; R + n.$$

If $R = 0$ **(equal weights),** $f'(k) = n$: MPK equals population growth. This recovers the *golden rule* for stationary per-capita consumption.

Pareto Efficiency and the Golden-Rule Criterion

Total per-capita consumption (young + old, weighted by population):

$$c \;\equiv\; c^1 + \frac{c^2}{1+n}.$$

In steady state the resource constraint collapses to

$$f(k) - nk = c. \tag{4}$$

Differentiating:

$$\frac{dc}{dk} = f'(k) - n.$$

- $f'(k) > n$: lowering k *reduces* stationary consumption $\Rightarrow$ economy is **dynamically efficient**.
- $f'(k) < n$: lowering k *raises* stationary consumption $\Rightarrow$ economy is **dynamically inefficient**.

Is the Decentralised Equilibrium Efficient?

Decentralised steady state: $(1+n)k = s(w(k), r(k))$.

Cobb-Douglas + **log utility**:

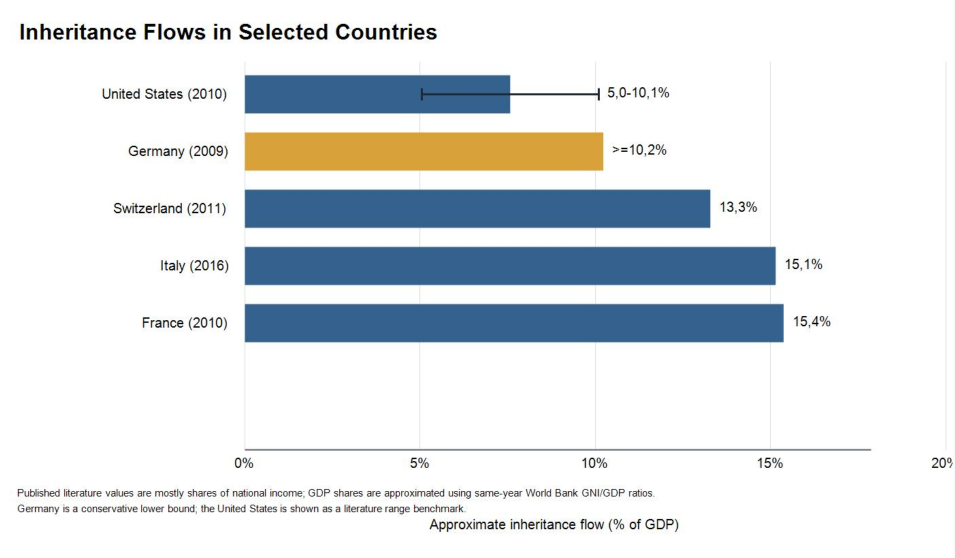
$$k_{t+1} = \frac{s_t}{1+n} = \frac{\beta}{1+\beta}\frac{1-\alpha}{1+n}k_t^{\alpha} \;\Rightarrow\; k = \left(\frac{\beta}{1+\beta}\frac{1-\alpha}{1+n}\right)^{1/(1-\alpha)}.$$

In this steady state,

$$r = f'(k) \gtreqless n \quad \Longleftrightarrow \quad \boxed{\frac{\alpha}{1-\alpha}\frac{1+\beta}{\beta} \gtreqless \frac{n}{1+n}}$$

With $\alpha \approx 0.36$, $\beta \approx 0.40$ (30-yr), $n = 0.10$:
$\frac{\alpha}{1-\alpha}\frac{1+\beta}{\beta} = \frac{0.36}{0.64} \cdot \frac{1.4}{0.4} \approx 1.97$, whereas $\frac{n}{1+n} \approx 0.09$.
The LHS dwarfs the RHS $\Rightarrow$ **Pareto inefficiency is unlikely**. Getting $f'(k) < n$ would require implausibly high β or low α.

Altruism: Restoring the Command Optimum



Altruism: Restoring the Command Optimum

Parent's recursive utility (values children's lifetime utility at rate R):

$$V_t = u(c_t^1) + \beta u(c_{t+1}^2) + \frac{1}{1+R} V_{t+1}.$$

Solving it forward:

$$V_t = \sum_{i=0}^{\infty} \frac{1}{(1+R)^i} \left[u(c_{t+i}^1) + \beta u(c_{t+i+1}^2) \right].$$

This is *exactly* the planner's objective (up to the initial-old term). Altruism turns the dynasty into an infinitely-lived decision-maker.

The Altruistic Household's Problem

Budget with *non-negative* bequests $beq_t \geq 0$:

$$c_t^1 + s_t = w_t + beq_t,$$

$$c_{t+1}^2 + (1+n)beq_{t+1} = (1+r_{t+1})s_t.$$

(Negative bequests would mean leaving debt to children: ruled out.) Intertemporal form:

$$c_t^1 + \frac{c_{t+1}^2 + (1+n)beq_{t+1}}{1+r_{t+1}} = w_t + beq_t.$$

Maximising the obj. fct. s.t. intertemporal constraint:

$$u'(c_t^1) = \beta\left[1 + r_{t+1}\right]u'(c_{t+1}^2),$$

$$u'(c_t^1)\,\frac{1+n}{1+r_{t+1}}\begin{cases} = \dfrac{u'(c_{t+1}^1)}{1+R} & \text{if } beq_{t+1} > 0, \\ \geq \dfrac{u'(c_{t+1}^1)}{1+R} & \text{if } beq_{t+1} = 0. \end{cases}$$

Equivalence with the Command Optimum

Result (Barro-style equivalence)

If bequests are **interior** ($beq_{t+1} > 0$) and factor prices clear at $r_{t+1} = f'(k_{t+1})$, the altruistic household's FOCs coincide with the planner's FOCs: $\Rightarrow$ the decentralised economy *is* Pareto-efficient.

When does the equivalence break?

- **Corner** $beq_{t+1} = 0$: the non-negativity constraint binds; parents would *like* to "take from children" but cannot.
- Empirical evidence (Altonji–Hayashi–Kotlikoff 1997): implications of altruism for intergenerational *risk sharing* are rejected in the data. The operative-bequest assumption is strong.

Numerical Method for Command Optimum

Use the Python Code **diamond.ipynb** to replicate this analysis.

The planner's problem under log utility and $R = 0$ reduces (using $c_t^2 = \beta(1+n)c_t^1$ from) to a **first-order system in the state-control pair** (k_t, c_t^1):

$$k_{t+1} = \frac{k_t + k_t^{\alpha} - (1+\beta)c_t^1}{1+n},$$
$$c_{t+1}^1 = c_t^1 \, \frac{1 + \alpha \, k_{t+1}^{\alpha-1}}{1+n}. \tag{5}$$

k_t is a *state* (given by history), c_t^1 is a *jump variable* (chosen by the planner).

Given k_0, *which* c_0^1 should the planner choose? Not all values are admissible — only those consistent with the TVC $\lim_{t\to\infty}(1+R)^{-t}\lambda_t k_{t+1} = 0$.

Saddle-Path Structure via Linearisation

Linearise the planner's system around the steady state (k^*, c^{1*}) where $k^* = (\alpha/n)^{1/(1-\alpha)}$ and $c^{1*} = [f(k^*) - nk^*]/(1+\beta)$:

$$\begin{pmatrix} k_{t+1} - k^* \\ c^1_{t+1} - c^{1*} \end{pmatrix} = J \begin{pmatrix} k_t - k^* \\ c^1_t - c^{1*} \end{pmatrix}, \qquad J = \begin{pmatrix} 1 & -\frac{1+\beta}{1+n} \\ \frac{c^{1*} f''(k^*)}{1+n} & 1 + \frac{c^{1*} f''(k^*)}{1+n}\left(-\frac{1+\beta}{1+n}\right) \end{pmatrix}.$$

For our calibration ($\alpha = 0.36, \beta = 0.40, n = 0.10$):

$$J \approx \begin{pmatrix} 1 & -1.273 \\ -0.0074 & 1.009 \end{pmatrix}, \quad \det J = 1, \quad \operatorname{tr} J \approx 2.009.$$

The eigenvalues

$$\mu_s = 0.9076 < 1 < 1.1018 = \mu_u, \qquad \mu_s \cdot \mu_u = 1.$$

Saddle $\Rightarrow$ unique stable manifold.

The TVC-compatible initial condition. Choose c^1_0 so that $(k_0 - k^*, c^1_0 - c^{1*})$ lies along the *stable eigenvector* v_s. Any other choice places weight on the unstable direction and violates the TVC.

Computing the Saddle Path: Eigenvector Projection

Step 1. Find v_s (eigenvector of J for $\mu_s < 1$ that is $(J - \mu_s I)\nu_s = 0$). Its *slope* gives the saddle-path policy rule:

$$s \equiv \frac{v_s^{(c)}}{v_s^{(k)}} \;\Rightarrow\; c^{1,\mathrm{SP}}(k) = c^{1*} + s\,(k - k^*).$$

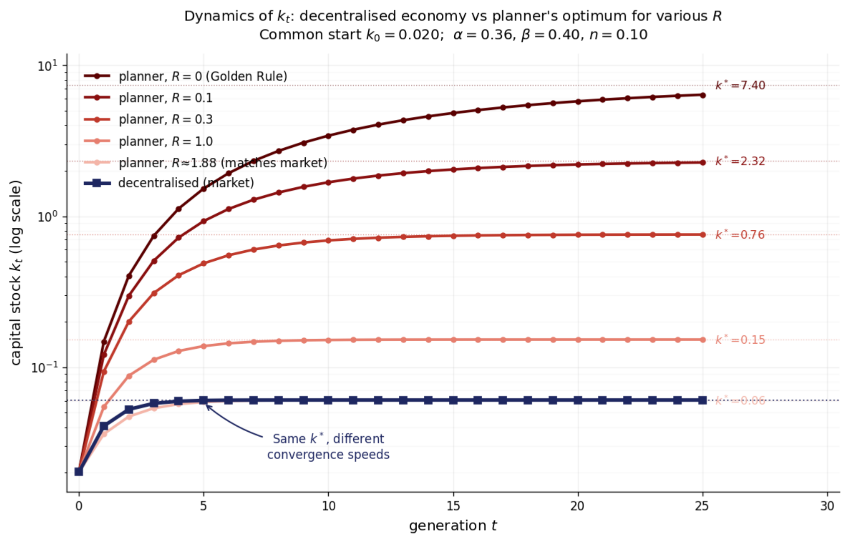
Step 2. For our calibration, $s = 0.0726$ (young consumption rises by ≈ 0.07 units for each unit rise in k along the saddle path).

Step 3. Given $k_0 = 0.7k^* = 5.18$:

$$c_0^1 = c^{1*} + 0.0726\,(k_0 - k^*) = 0.9397 + 0.0726 \cdot (-2.22) = 0.7785.$$

Step 4. Iterate the nonlinear system from $(k_0, c_0^1) = (5.180, 0.779)$. Path converges to (k^*, c^{1*}) at rate $\mu_s = 0.91$ per generation.

Comparison Decentralised vs Planner's OLG



Decentralised vs Planner's OLG: The Punchline

	Decentralised OLG	Planner's OLG
Dynamic system	1D: $k_{t+1} = g(k_t)$	2D: (k_t, c_t^1)
Eigenvalues	one: $\alpha = 0.36$	two: $0.91, 1.10$
Jump variable	none	c_t^1
TVC	absent	restored
Saddle path	trivial	non-trivial
Convergence persistence	0.36 per gen.	0.91 per gen.
Who optimises long-run?	no one	planner

Summary

Summary: Key Results — Lecture 2

1. **Exchange $\to$ production.** Euler equation identical. $w_t = \omega(k_t)$, $r_t = \rho(k_t)$ endogenous.
2. $s(w, r)$**.** $\partial s/\partial w \in (0, 1)$ always. $\partial s/\partial r \gtrless 0 \Leftrightarrow$IES$\gtrless 1 \Leftrightarrow$ backward-bending offer curve (L1) $\Leftrightarrow$ poverty traps (L2).
3. **Offer curve in** (c^1, c^2) **space.** Budget slope $-(1 + r^*)$ vs. feasibility slope $-(1 + n)$. Samuelson: $r^* < n \Leftrightarrow$ budget shallower than feasibility. Golden Rule $(r^* = n)$ is the L2 analog of the L1 monetary equilibrium.
4. **Transition map** $g(k_t)$**.** Unique stable SS in CD+log $(g' = \alpha < 1)$. Non-monotone $\Rightarrow$ poverty traps (IES< 1).
5. $r^* \geq n \Leftrightarrow$ **dynamic efficiency.** Production counterpart of L1's $\bar{r} < 0$ (Samuelson).
6. **PAYG** $=$ **Social Ponzi (L1).** Return $1 + n$ replaces $1 + \bar{r}$. Dominates iff $r^* < n$.
7. $r < g$**:** impossible in Ramsey (TVC), possible in OLG (no aggregate TVC).

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